

Technology Stack Analysis

MERN + Socket.io + Vite Stack Selection Rationale

1. Core Architectural Stack

- **MongoDB (v7.0.8):** Document-oriented database storing profiles and nested slot matrices. Highly queryable for date coordinates and indexes.
- **Express.js (v4.18):** Routing middleware handling incoming patient API requests, security checks, and cross-dashboard data updates.
- **React.js (Vite v8):** Frontend single page app compiling quickly. Virtual DOM renders schedule inputs instantly without reload delays.
- **Node.js (v22.13.0):** JavaScript runtime offering non-blocking asynchronous event loop handles multi-user connections.

2. Utility Packages and Integrations

- **Socket.io:** Bi-directional TCP socket communication. Emits immediate alerts to connected users.
- **Tailwind CSS:** Premium custom glassmorphic aesthetics. Adaptive CSS grid configurations.
- **Chart.js:** Visualized statistics diagrams tracking booking growth on the Admin Panel.